Supplementary Material: A Validated Leakage Audit for Wearable Physiological Regression, with Application to PPG-Based Glucose and Blood-Pressure Estimation

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S1. TRIPOD+AI checklist

Items of the TRIPOD+AI statement [1] and where each is addressed (main paper: Sec. I–VI; supplement: S2–S8). Items that do not apply to an evaluation of public data are marked n/a.

Item	Requirement	Where
1	Title identifies study as developing/evaluating prediction models	Title, Abstract
2	Abstract: objectives, data, methods, results, conclusions	Structured abstract
3a–b	Background, rationale, objectives	Sec. I
4	Data sources, dates, setting	Sec. IV-A; S3
5a	Participants: eligibility, recruitment	Sec. IV-A, S3 (as reported by dataset authors)
5b–c	Treatment / care pathway	n/a
6	Data preparation: cleaning, windowing, missing data	Sec. IV-A; S2; loaders in code
7	Outcome definition and measurement	Sec. IV-A; S3 (fingerstick meter; Dexcom G6; cuff BP)
8	Predictors: definition and measurement	Sec. IV-A; S2
9	Sample size justification	Sec. III (power of the decision rule); S5
10	Missing-data handling	S2 (gap-free windows only)
11	Analytical methods: model type, hyper-parameters, selection	Sec. IV-A; S2
12a	Evaluation: splits, protocol, leakage control	Sec. II-C, II-D; Table I; frozen folds in code
12b	Model performance measures	Sec. II-E
12c	Model comparison and null models	Sec. II-E; S8
12d	Uncertainty (CIs, tests)	Sec. II-E; Tables II–III; S6, S8
12e	Fairness / subgroup analysis	Per-subject results (Fig. S3b); sex not analysed (n too small)
13	Class imbalance / outcome distribution	S3; glucose ranges and hypo-/hyperglycaemic counts
14	Open science: code, data, protocol	Title footnote; README (incl. encoder weight sources and licences)
15	Patient and public involvement	n/a (public data)
16–17	Participant flow, characteristics	Sec. IV-A; S3
18a–f	Model performance, comparisons, uncertainty, subgroups	Sec. IV-B, IV-C; S6–S8
19	Model updating	n/a
20	Interpretation; comparison with prior work	Sec. V; Table S6
21	Limitations	Sec. V
22	Usability: inputs, outputs, intended use	S2 (intended use; nulls per use case in Sec. II-E)
23	Funding, conflicts	To be completed by the author

S2. Model information sheet

Field	Value
Task	Regression of blood / interstitial glucose (mg/dL) from a 5-s PPG window (networks, classical) or 10–30-s segments (frozen encoders); systolic and diastolic blood pressure (mmHg) from 2.1-s PPG-BP segments
Intended use	Benchmark only; no clinical use
Inputs	320 samples at 64 Hz, 0.5–8 Hz bandpassed, z-scored per window (variants in Table S12)
Outputs	One glucose value per window; averaged per reference measurement
Training data	Kermani–Esmaeili recordings (LOSO folds); BIG IDEAs training periods; STUBA ePPG (LOSO); PPG-BP (5-fold subject-wise and random-segment folds)
Hyper-parameters	AdamW, lr 10^{-3}, wd 10^{-4}, one-cycle, 40/15 epochs, batch 16/64, grad-clip 1, smooth-L1 on standardised target
Seeds	Kermani–Esmaeili networks: 3; STUBA 1-D CNN: 3, other STUBA networks one; BIG IDEAs residual 1-D CNN: 3, other BIG IDEAs rows one; encoder probes and classical models are deterministic (one run)
Null models	Train mean; subject train mean; time of day; persistence
Leakage controls	Named protocols with asserted guarantees; frozen folds; guard gap; label-construction detector, whole-column and block scan
Known failure modes	Predicts near the (subject) mean; no event sensitivity
Pretrained encoders	PaPaGei-S (5.8 M, 125 Hz, 10-s), Pulse-PPG (28.5 M, 50 Hz, 10–30-s), AnyPPG (4.0 M, 125 Hz, 10-s); frozen released weights loaded weights-only; 0.5–12 Hz Chebyshev-II, resample, z-score; 512-d embeddings averaged per reference; RidgeCV ($\alpha \in [10^{-2}, 10^{4}]$) on training-fold-standardised features
Licences	Data: Kermani–Esmaeili CC BY 4.0 (Mendeley Data); BIG IDEAs PhysioNet ODC-By; STUBA ePPG CC0; PPG-BP CC0 (figshare). Encoders: PaPaGei code BSD-3-Clause, weights Zenodo 13983110 (no licence stated); Pulse-PPG MIT (code and weights, Zenodo 17345536); AnyPPG checkpoint from GitHub PKUDigitalHealth/AnyPPG (no licence file)

S3. Cohort characteristics

	Kermani–Esmaeili	**BIG IDEAs**	**STUBA ePPG**	**PPG-BP**
Subjects	23	16	24	219
Male / female	15 / 8	7 / 9	20 / 4	see [2]
Age (years)	22–61	35–65 (inclusion criterion)	22–48	21–86
Diabetes	not stated	HbA1c 5.3–6.4%	3 T1DM	not an outcome
Reference	Accu-Chek Active	Dexcom G6 CGM	Beurer GL49	cuff (Omron HEM-7201)
Reference measurements	67	28867	125	219 (657 segments)
Glucose range (mg/dL)	88–183	40–261	40–304	–
< 70 / > 180 mg/dL	0 / 2	186 / 431	34 / 18	–
Sensor	finger, 550 nm, 2175 Hz	wrist E4 BVP, 64 Hz	finger, 530/655/940 nm, 100 Hz	finger, 1000 Hz, 2.1-s segments

S4. Validation of the label-construction detector

Synthetic label columns of 20 subjects, 200 rows per reference value; a corrupted subject has a contiguous fraction of its rows replaced by the cross-join pattern (its k reference values cycled row by row). Thirty replicates per condition. The detector is `detect_label_cycling` (whole column) and `detect_label_cycling_local` (block scan, $b = 256$, stride 128, $\tau = 0.98$, $\ell_{\max} = 32$) in `ppgbench.integrity`; the experiment is `validate_detectors.py`. Residual misses at $k = 2$ are draws in which a subject's two reference values coincide, so that the column contains no cycle to detect.

TABLE S3

Cross-join detector on synthetic label columns: subject-level sensitivity (%) by cycle length k and fraction of a corrupted subject's rows that cycle, pooled over 10–100 % corrupted subjects. Specificity: 100 % (whole-column) and 100 % (block scan) in every condition.

	Whole-column			**Block scan**		
k	25 %	50 %	100 %	25 %	50 %	100 %
2	99	98	98	99	98	98
3	100	100	100	100	100	100
5	100	100	100	100	100	100
7	0	100	100	100	100	100
10	0	4	100	100	100	100

Real negative control, Kermani–Esmaeili: 0 / 0 of 23 subjects flagged (whole-column / block)
Real negative control, STUBA ePPG: 0 / 0 of 24 subjects flagged (whole-column / block)
Real negative control, BIG IDEAs: 0 / 0 of 16 subjects flagged (whole-column / block)
Real positive control, Kaggle derivative: 21 / 21 of 21 patients with > 1 reference value flagged

S5. Validation of the null-anchored decision rule

Each synthetic cohort has 16 subjects $\times$ 12 records $\times$ three overlapping 5-s windows generated by a pulse model with heart-rate variability, respiration and noise. Label = subject offset ($\mathcal{N}(120, 20^2)$, not encoded in the signal) $+ g\,(\mathrm{HR} - 70)$ $+ \mathcal{N}(0, 3^2)$, with within-subject HR SD 8 bpm. Each subject has a label-independent pulse-shape fingerprint (second-harmonic amplitude $\sim U(0.15, 0.75)$, notch amplitude $\sim U(0.05, 0.45)$). Models use the nine hand-crafted features of the main paper; predictions are averaged per record and scored with 2 000 subject-cluster bootstrap draws.

TABLE S4

Null-anchored decision rule on synthetic cohorts (200 per condition): percentage of cohorts in which the model is declared better than the train-mean null (95 % subject-cluster interval of ΔMAE entirely below zero), and the mean ΔMAE (mg/dL). g: true learnable signal in mg/dL per bpm; at $g = 0$ every claim is false (nominal one-sided rate 2.5 %).

Protocol	**Model**	$g = 0$	$g = 1$	$g = 2$	$g = 3$	$g = 4$
subject-wise (honest)	ridge	1.0 % (+1.7)	5.5 % (+0.1)	31.5 % (-4.4)	80.0 % (-11.3)	96.5 % (-18.7)
subject-wise (honest)	GBM	1.0 % (+3.1)	2.0 % (+1.7)	20.5 % (-2.9)	69.0 % (-10.3)	93.0 % (-17.3)
random windows (leaky)	ridge	19.0 % (-1.9)	53.0 % (-3.9)	95.5 % (-8.6)	100.0 % (-15.3)	100.0 % (-22.3)
random windows (leaky)	GBM	100.0 % (-9.3)	100.0 % (-11.6)	100.0 % (-16.5)	100.0 % (-23.3)	100.0 % (-30.1)

S6. PPG-BP: full results

S7. PPG $\rightarrow$ glucose: detailed results

Tables and figures that support Sec. IV-B of the main paper. The fusion network is shown in Fig. S1; published results on the Kermani–Esmaeili recordings are in Table S6; the full leakage ladder is in Table S7.

TABLE S5

PPG-BP, all rows. MAE and RMSE in mmHg; networks mean ± std over seeds; Δ: paired MAE difference against the train-mean null with its 95 % subject-cluster interval (most conservative interval over seeds).

Target	Split	Model	MAE	RMSE	r	Δ [95% CI]
SBP	random segments	Train mean (null)	16.3	20.4	-0.16	–
SBP	random segments	Ridge, demographics	13.7	18.0	0.47	-2.6 [-3.65, -1.51]
SBP	random segments	Ridge, PPG features	15.0	19.1	0.35	-1.3 [-1.84, -0.72]
SBP	random segments	Ridge, PPG + demogr.	13.6	17.5	0.51	-2.7 [-3.71, -1.65]
SBP	random segments	GBM, PPG + demogr.	9.4	11.9	0.82	-6.8 [-8.28, -5.53]
SBP	random segments	1-D CNN	12.2±0.2	15.7±0.2	0.64±0.01	-4.1 [-5.14, -2.55]
SBP	subject-wise	Train mean (null)	16.2	20.4	-0.12	–
SBP	subject-wise	Ridge, demographics	13.9	18.3	0.44	-2.4 [-3.47, -1.30]
SBP	subject-wise	Ridge, PPG features	15.1	19.3	0.31	-1.1 [-1.73, -0.55]
SBP	subject-wise	Ridge, PPG + demogr.	13.8	18.0	0.47	-2.4 [-3.48, -1.35]
SBP	subject-wise	GBM, PPG + demogr.	14.6	18.6	0.44	-1.7 [-3.13, -0.30]
SBP	subject-wise	1-D CNN	13.4±0.1	17.2±0.1	0.54±0.01	-2.9 [-4.21, -1.36]
DBP	random segments	Train mean (null)	8.7	11.1	-0.06	–
DBP	random segments	Ridge, demographics	8.4	10.7	0.26	-0.3 [-0.65, +0.03]
DBP	random segments	Ridge, PPG features	8.6	10.9	0.17	-0.1 [-0.25, +0.09]
DBP	random segments	Ridge, PPG + demogr.	8.1	10.4	0.35	-0.6 [-1.01, -0.18]
DBP	random segments	GBM, PPG + demogr.	5.5	7.2	0.78	-3.2 [-3.89, -2.48]
DBP	random segments	1-D CNN	7.2±0.1	9.3±0.2	0.55±0.03	-1.5 [-2.02, -0.72]
DBP	subject-wise	Train mean (null)	8.7	11.1	-0.05	–
DBP	subject-wise	Ridge, demographics	8.5	10.8	0.23	-0.2 [-0.53, +0.16]
DBP	subject-wise	Ridge, PPG features	8.7	11.1	0.07	+0.0 [-0.16, +0.19]
DBP	subject-wise	Ridge, PPG + demogr.	8.4	10.6	0.30	-0.4 [-0.81, +0.07]
DBP	subject-wise	GBM, PPG + demogr.	8.2	10.4	0.38	-0.6 [-1.25, +0.12]
DBP	subject-wise	1-D CNN	7.9±0.1	10.2±0.1	0.42±0.01	-0.8 [-1.39, -0.05]

TABLE S6

Published results on the Kermani–Esmaeili recordings (23 subjects, 67 fingerstick references) and the null-anchored result of this paper. MAE in mg/dL.

Study	Model	Split	MAE
Soliman 2024 [3]	CNN–GRU	augment, then shuffle	2.96
Kermani & Esmaeili 2025 [4]	SDF-MFCC + SVR	5-fold, level unstated	4.0†
Syamsul & Kim 2026 [5]	attention CNN–GRU	subject-level (stated)	11.59
Syamsul & Kim 2026 [5]	1-D CNN baseline	subject-level (stated)	13.45
Zeynali 2025 [6]	ResNet-34, VitalDB-trained	external test	14.8
Ramesh 2026 [7]	train-mean null, VitalDB	external test	16.0
This paper	train-mean null	LOSO, recording level	14.7
This paper	1-D CNN, 3 seeds	LOSO, recording level	13.4±0.5

†reported together with RMSE 26.0 and MARD 0.195%, which are mutually inconsistent.

TABLE S7

Leakage ladder, 1-D CNN and nulls; MAE in mg/dL. Null: train mean for the Kermani–Esmaeili recordings and the Kaggle derivative; each subject's own training-period mean for BIG IDEAs, which coincides with the leaky subject mean. K–E window, recording and LOSO rungs: three seeds (mean ± std, the LOSO rung being the experiment of Table S8); all other rungs: one seed. The BIG IDEAs random-readings rung is scored on all 28867 pooled out-of-fold readings, the chronological rungs on the 8664 held-out readings.

Cohort	Protocol	Null	Leaky subj. mean	1-D CNN
K–E	random windows	14.4	12.3	10.4±0.2
	random recordings	14.7	17.8	13.5±0.5
	subject-wise (LOSO)	14.7	–	13.4±0.5
Kaggle deriv.	random windows	9.7	0.03	9.5
	subject-wise (LOSO)	10.2	–	10.2
BIG IDEAs	random readings	15.0	15.0	16.1
	chronological, no gap	15.2	–	16.9
	chronological, 1-h gap	15.3	–	16.9

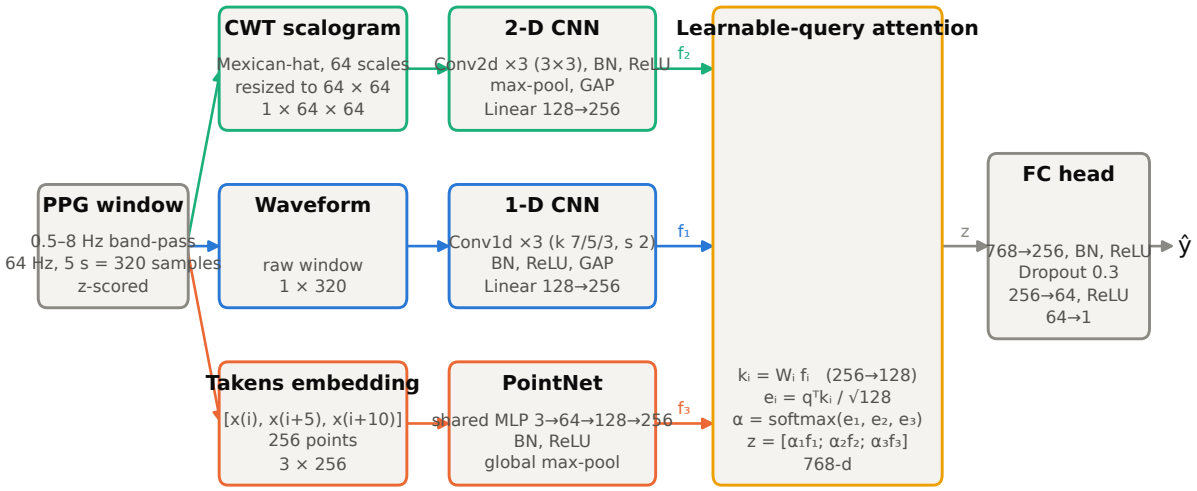


Fig. S1. The 1D–2D–3D fusion network evaluated as one row of the benchmark. One PPG window is presented as a waveform, a wavelet scalogram and a Takens phase-space point cloud; three equal-width encoders feed a learnable-query attention that weights the branches before a regression head.

TABLE S8

LEAVE-ONE-SUBJECT-OUT RESULTS ON THE KERMANI–ESMAEILI RECORDINGS (67 RECORDINGS, 23 SUBJECTS, RECORDING LEVEL). MAE AND RMSE IN MG/DL; A AND A+B ARE CLARKE FRACTIONS IN %. NETWORKS: MAE, RMSE, r AND ZONES ARE MEAN ± STD OVER 3 SEEDS; Δ AND ITS 95% SUBJECT-CLUSTER BOOTSTRAP INTERVAL ARE COMPUTED ON THE SEED-ENSEMBLE PREDICTION (MEAN OF THE SEEDS' PREDICTIONS PER RECORDING), WHOSE MAE IS GIVEN IN PARENTHESES, AGAINST THE TRAIN-MEAN NULL. THE PRIMARY COMPARISON IS MARKED ⋆; OTHER ROWS ARE EXPLORATORY.

	Model	MAE	RMSE	r	A	A+B	Δ vs. null [95% CI]
baseline	Train mean (null)	14.7	18.9	-0.59	83.6	100.0	–
	Reference-noise floor (ISO 15197)	7.1	–	–	–	–	–
	Ridge, demographics	15.6	19.7	-0.17	86.6	100.0	+0.9 [-0.3, +2.3]
	Ridge, PPG features	14.6	18.9	-0.03	86.6	100.0	-0.1 [-0.6, +0.6]
	Gradient boosting, PPG + demogr.	15.5	20.6	-0.05	80.6	100.0	+0.8 [-1.3, +3.0]
network	1-D CNN ⋆	13.4±0.5 (13.3)	18.0±0.2	0.32±0.02	80.6±1.2	100.0±0.0	-1.4 [-3.0, -0.03]
	2-D CNN (CWT)	13.9±0.2 (13.9)	18.5±0.1	0.22±0.01	84.1±0.7	100.0±0.0	-0.8 [-2.5, +0.7]
	PointNet (Takens)	14.0±0.1 (14.0)	18.6±0.1	0.14±0.03	88.1±0.0	100.0±0.0	-0.7 [-1.8, +0.4]
	ResNet-1D [6]	15.1±0.4 (15.0)	19.2±0.5	0.19±0.05	80.6±1.2	100.0±0.0	+0.3 [-1.4, +2.0]
	CNN–LSTM [5]	13.5±0.2 (13.3)	18.7±0.3	0.19±0.02	85.6±0.7	100.0±0.0	-1.4 [-2.9, +0.2]
	1D+2D+3D cross-attention	14.8±0.3 (14.7)	19.3±0.4	0.13±0.05	78.6±0.7	100.0±0.0	-0.1 [-1.5, +1.2]
probe	PaPaGei-S + ridge [8]	14.7	19.1	-0.01	83.6	100.0	+0.0 [-0.9, +1.0]
	Pulse-PPG + ridge [9]	14.2	18.5	0.14	80.6	100.0	-0.5 [-1.7, +0.8]
	AnyPPG + ridge [10]	14.9	19.2	-0.33	85.1	100.0	+0.2 [-0.2, +0.7]

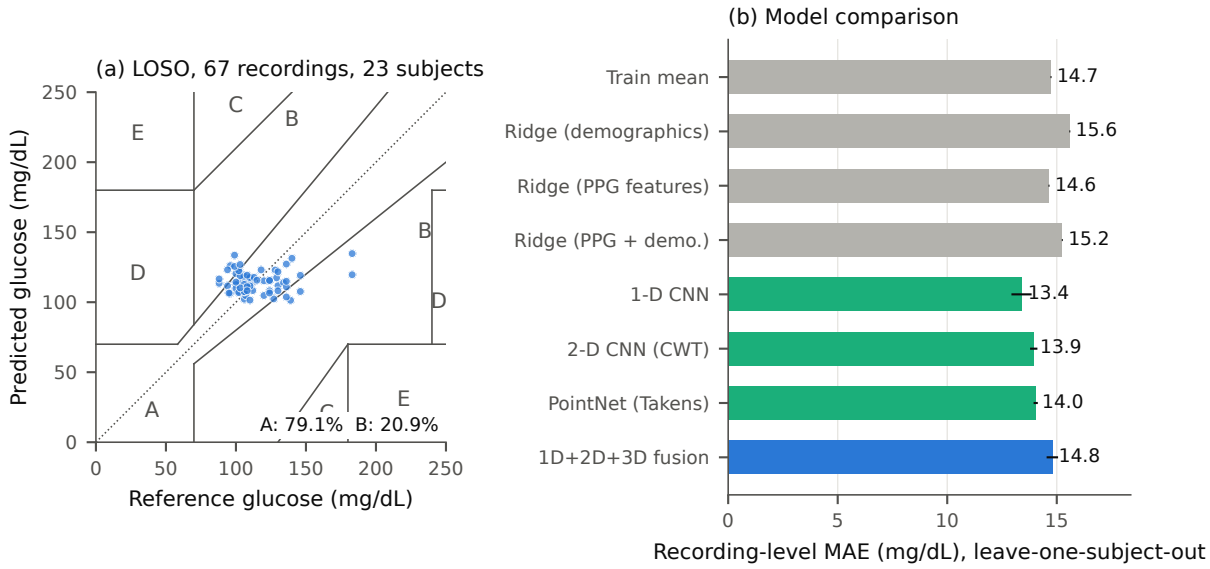


Fig. S2. Leave-one-subject-out results on the Kermani–Esmaeili recordings. (a) Clarke error grid of the fusion network's out-of-fold recording-level predictions: the predictions occupy a narrow band around the training mean. (b) Recording-level MAE of every model; grey bars are baselines that do not see the PPG waveform through a network.

TABLE S9

WITHIN-SUBJECT CHRONOLOGICAL SPLIT WITH 1-H GUARD GAP ON BIG IDEAS (8664 HELD-OUT CGM READINGS, 16 SUBJECTS). MAE AND RMSE IN MG/DL, MARD IN %; r_w: MEDIAN WITHIN-SUBJECT CORRELATION. Δ: PAIRED MAE DIFFERENCE AGAINST THE SUBJECT-MEAN NULL WITH ITS 95% SUBJECT-CLUSTER BOOTSTRAP INTERVAL, COMPUTED ON THE READINGS BOTH ROWS PREDICT. THE ENCODER PROBES EXCLUDE 7 OF 28867 READINGS THAT HAVE NO 10-S SEGMENT PASSING THE QUALITY CHECK, WHICH MOVES THE CHRONOLOGICAL CUT SO THAT 4 OF THE 8664 HELD-OUT READINGS DIFFER (PROBE ROWS ARE SCORED ON 8663 READINGS). 1-D CNN, RESIDUAL: MEAN $\pm$ STD OVER 3 SEEDS; ALL OTHER ROWS ONE RUN.

	Model	MAE	RMSE	MARD	r	r_w	Δ [95% CI]
null	Subject train mean	15.3	21.6	12.8	0.43	–	–
	Reference-noise floor (MARD 9%)	10.5	–	9.0	–	–	–
	Time of day + subject	15.3	21.8	12.8	0.41	-0.01	+0.0 [-0.5, +0.5]
	Persistence	31.3	44.9	27.3	0.21	–	+16.1 [+3.8, +29.9]
population	Ridge, PPG features	17.3	23.8	14.9	0.05	0.05	+2.0 [+0.7, +3.8]
	1-D CNN	16.9	23.8	14.2	0.21	0.04	+1.7 [+0.5, +3.1]
	2-D CNN (CWT)	17.0	23.8	14.3	0.19	0.03	+1.7 [+0.5, +3.2]
	PointNet (Takens)	17.0	23.8	14.3	0.20	0.03	+1.7 [+0.5, +3.2]
	1D+2D+3D cross-attention	16.7	23.5	14.0	0.25	0.03	+1.4 [+0.2, +2.9]
personal.	Ridge, residual	15.1	21.6	12.7	0.44	0.06	-0.1 [-0.4, +0.1]
	1-D CNN, residual	15.0±0.0	21.9±0.0	12.3±0.0	0.44±0.00	0.04±0.00	-0.2 [-0.7, +0.2]
	Fusion, residual	15.1	21.9	12.3	0.44	0.07	-0.2 [-0.7, +0.3]
probe	PaPaGei-S + ridge	17.2	23.8	14.8	0.07	0.03	+2.0 [+0.6, +3.7]
	Pulse-PPG + ridge	17.0	23.6	14.6	0.15	0.04	+1.7 [+0.4, +3.4]
	AnyPPG + ridge	16.8	23.3	14.4	0.23	0.07	+1.5 [+0.3, +3.2]
	PaPaGei-S, residual	15.1	21.6	12.6	0.43	0.03	-0.2 [-0.5, +0.04]
	Pulse-PPG, residual	15.0	21.5	12.5	0.44	0.07	-0.3 [-0.5, -0.02]
	AnyPPG, residual	15.1	21.4	12.6	0.45	0.17	-0.2 [-0.6, +0.2]

TABLE S10

STUBA EPPG, LEAVE-ONE-SUBJECT-OUT, BY CHANNEL; MAE IN MG/DL. NULL: TRAIN MEAN. 1-D CNN: MEAN $\pm$ STD OVER 3 SEEDS; OTHER CELLS ONE RUN. PAIRED INTERVALS FOR THE IR1 ROWS ARE GIVEN IN TABLE II OF THE MAIN PAPER AND ALL COMPARISONS IN SEC. S8.

Channel	Null	Ridge	GBM	1-D CNN
940 nm, site 1 (IR1)	43.5	44.4	48.3	49.7±1.3
940 nm, site 2 (IR2)	43.5	44.9	45.7	–
655 nm, site 1 (R1)	43.5	43.9	49.3	–
655 nm, site 2 (R2)	43.5	44.9	47.7	–
530 nm, site 1 (G1)	43.5	44.1	47.9	–
530 nm, site 2 (G2)	43.5	44.3	46.8	–
six-channel average	43.5	44.1	46.8	–

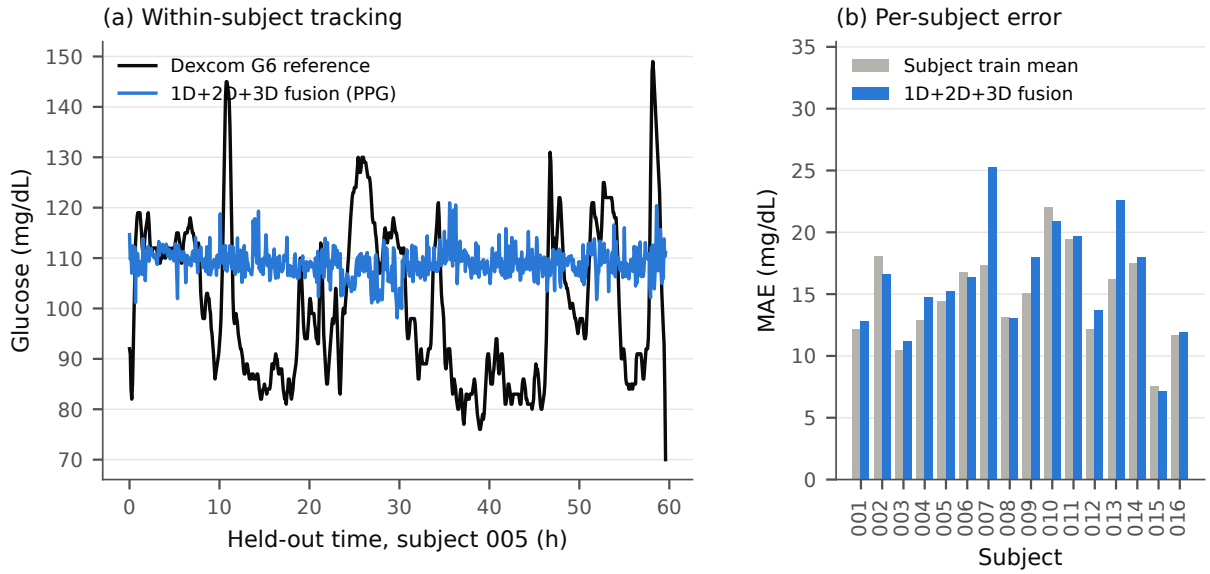


Fig. S3. Within-subject tracking on BIG IDEAs. (a) Dexcom G6 reference and prediction of the population 1D–2D–3D fusion network over the held-out final 30% of one subject's recording. (b) Per-subject MAE of the model against each subject's own training-period mean.

TABLE S11

CONTROLS THROUGH THE IDENTICAL PIPELINE AND PROTOCOL (1-D CNN; KERMANI–ESMAEILI ROWS MEAN ± STD OVER THREE SEEDS, BIG IDEAS ROW ONE SEED). MAE IN TARGET UNITS; r: PEARSON.

Cohort	Target	Expected	MAE	Null	r
K–E	heart rate (bpm)	recoverable	20.5±0.3	41.9±0.0	0.74±0.01
K–E	age (years)	weak / none	8.9±0.1	8.1±0.0	-0.09±0.01
K–E	glucose	(task)	13.4±0.5	14.7±0.0	0.32±0.02
K–E	glucose, permuted	null	15.6±0.3	14.6±0.0	-0.10±0.13
K–E	glucose, noise input	null	14.6±0.4	14.7±0.0	-0.07±0.09
BIG IDEAs	heart rate (bpm)	recoverable	9.7	14.2	0.63

TABLE S12

REPRESENTATION SENSITIVITY ON BIG IDEAS (GUARDED CHRONOLOGICAL SPLIT, RESIDUAL TARGETS). MAE IN MG/DL; r_w: MEDIAN WITHIN-SUBJECT CORRELATION. ONE WINDOW PER READING AND ONE SEED PER ROW (TABLE S9 USES TWO WINDOWS AND THREE SEEDS, HENCE THE SLIGHTLY DIFFERENT DEFAULT ROW); LONGER WINDOWS KEEP ONLY READINGS WITH A GAP-FREE WINDOW OF THAT LENGTH.

Representation	Null	Ridge	r_w	1-D CNN	r_w
5 s, 0.5–8 Hz, z-score (default)	15.3	15.1	0.04	15.1	0.02
10 s	15.3	15.1	0.05	15.1	0.02
30 s (+ HRV features)	15.3	15.1	0.06	15.0	0.04
60 s (+ HRV features)	15.3	15.1	0.07	15.0	0.08
5 s, 0.1–12 Hz	15.3	15.1	0.03	15.1	0.02
5 s, amplitude retained	15.3	15.1	0.07	15.1	0.04
60 s, 0.1–12 Hz, amplitude, HRV	15.3	15.0	0.10	15.0	0.05

S8. All paired glucose comparisons against the null

Every model-versus-null comparison in the paper (41 in total; ⋆: the pre-specified primary comparison, all others exploratory). MAE of the (seed-ensemble) prediction in mg/dL; Δ: paired MAE difference against the cohort's null (train mean; subject training-period mean on BIG IDEAs) with its 95% subject-cluster bootstrap interval. "better"/"worse": interval entirely below/above zero. 2 intervals lie below zero and 16 above; no correction for multiplicity is applied.

Cohort	Model	MAE	Δ	95% CI	Verdict
Kermani–Esmaeili	Ridge, PPG	14.62	-0.09	[-0.64, +0.62]	–
Kermani–Esmaeili	Ridge, demographics	15.57	+0.86	[-0.32, +2.27]	–
Kermani–Esmaeili	Ridge, PPG + demogr.	15.23	+0.52	[-1.17, +2.57]	–
Kermani–Esmaeili	GBM, PPG	16.61	+1.89	[+0.15, +3.47]	worse
Kermani–Esmaeili	GBM, PPG + demogr.	15.49	+0.78	[-1.28, +2.99]	–
Kermani–Esmaeili	1-D CNN⋆	13.31	-1.41	[-2.99, -0.03]	better
Kermani–Esmaeili	2-D CNN (CWT)	13.91	-0.81	[-2.51, +0.75]	–
Kermani–Esmaeili	PointNet (Takens)	14.01	-0.70	[-1.83, +0.38]	–
Kermani–Esmaeili	1D+2D+3D fusion	14.66	-0.05	[-1.48, +1.25]	–
Kermani–Esmaeili	ResNet-1D	15.01	+0.30	[-1.42, +2.01]	–
Kermani–Esmaeili	CNN–LSTM	13.34	-1.37	[-2.95, +0.23]	–
Kermani–Esmaeili	PaPaGei-S probe	14.75	+0.03	[-0.91, +1.05]	–
Kermani–Esmaeili	AnyPPG probe	14.91	+0.19	[-0.17, +0.67]	–
Kermani–Esmaeili	Pulse-PPG probe	14.21	-0.50	[-1.67, +0.79]	–
BIG IDEAs	Ridge, PPG	17.32	+2.04	[+0.66, +3.83]	worse
BIG IDEAs	1-D CNN	16.94	+1.66	[+0.48, +3.11]	worse
BIG IDEAs	2-D CNN (CWT)	16.97	+1.69	[+0.50, +3.18]	worse
BIG IDEAs	PointNet (Takens)	16.98	+1.70	[+0.48, +3.22]	worse
BIG IDEAs	1D+2D+3D fusion	16.68	+1.40	[+0.22, +2.85]	worse
BIG IDEAs	Time of day + subject	15.29	+0.01	[-0.49, +0.53]	–
BIG IDEAs	Persistence	31.35	+16.07	[+3.83, +29.92]	worse
BIG IDEAs	Ridge, PPG, residual	15.13	-0.15	[-0.36, +0.09]	–
BIG IDEAs	1-D CNN, residual	15.04	-0.24	[-0.70, +0.23]	–
BIG IDEAs	2-D CNN (CWT), residual	15.09	-0.19	[-0.71, +0.31]	–
BIG IDEAs	PointNet (Takens), residual	15.04	-0.24	[-0.69, +0.17]	–
BIG IDEAs	1D+2D+3D fusion, residual	15.07	-0.21	[-0.73, +0.31]	–
BIG IDEAs	PaPaGei-S probe	17.24	+1.96	[+0.61, +3.68]	worse
BIG IDEAs	AnyPPG probe	16.79	+1.51	[+0.29, +3.20]	worse
BIG IDEAs	Pulse-PPG probe	16.99	+1.70	[+0.38, +3.41]	worse
BIG IDEAs	PaPaGei-S probe, residual	15.07	-0.21	[-0.46, +0.04]	–
BIG IDEAs	AnyPPG probe, residual	15.08	-0.21	[-0.58, +0.16]	–
BIG IDEAs	Pulse-PPG probe, residual	15.01	-0.27	[-0.54, -0.02]	better
STUBA ePPG	Ridge, PPG	44.39	+0.89	[+0.35, +1.22]	worse
STUBA ePPG	GBM, PPG	48.33	+4.83	[+1.76, +9.16]	worse
STUBA ePPG	1-D CNN	49.55	+6.05	[+2.35, +7.65]	worse
STUBA ePPG	2-D CNN (CWT)	48.61	+5.10	[+2.70, +6.32]	worse
STUBA ePPG	PointNet (Takens)	43.20	-0.30	[-1.28, +1.95]	–
STUBA ePPG	1D+2D+3D fusion	47.52	+4.01	[+1.30, +5.16]	worse
STUBA ePPG	PaPaGei-S probe	43.93	+0.43	[-0.01, +1.75]	–
STUBA ePPG	AnyPPG probe	43.51	+0.01	[-0.75, +1.08]	–
STUBA ePPG	Pulse-PPG probe	44.58	+1.08	[+0.07, +1.88]	worse

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